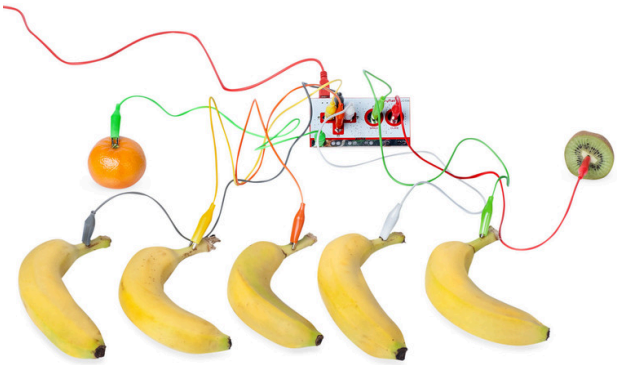
Element D: Design Concept Generation, Analysis, and Selection

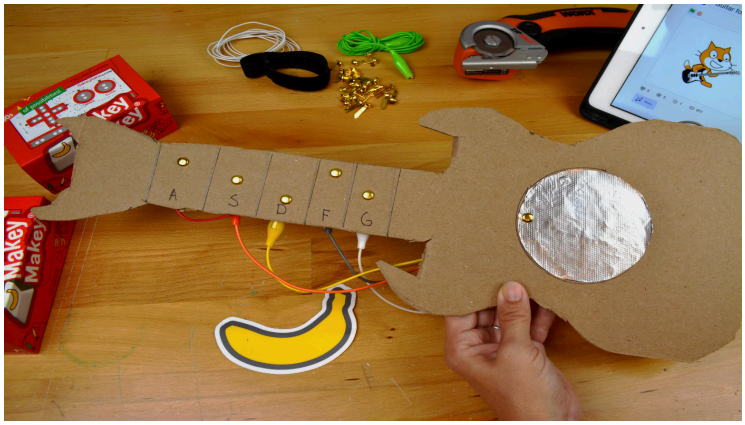
Week 1: Brainstorm and Concept Design

Our first project using Makey Makey is a banana piano, and it can be an exciting introduction to the world of Makey Makey. We will connect the alligator clips to bananas, which will serve as our piano keys. It will be interesting to show how the electricity travels through the bananas to make music. As we touch the bananas, they should complete the circuit and play different piano notes, creating a fun and engaging musical experience. This hands-on project will not only demonstrate the basic principles of conductivity and circuits but also allow us to explore our creativity by experimenting with different fruits and sounds. It should be a memorable way to start making things with Makey Makey, making learning fun and enjoyable.



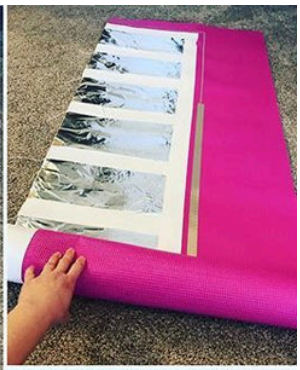
Week 2: The prototype

I'm excited about the idea of making a cardboard guitar with Makey Makey. I plan to design a simple guitar shape out of cardboard and use conductive materials for the strings, so when we strum them, they'll connect to the Makey Makey board and play music. By attaching alligator clips to the strings and linking them to different sounds, we can create a unique instrument that not only looks awesome but also produces a variety of musical notes. This project will be a great way to mix art and technology, and I can't wait to customize it with colors and decorations. Plus, it'll be great for the kids to see and try to make their own guitars, so we can all jam out and enjoy making music together!



Week 3: Prototype 3 and Test Trials

In Week 3, we plan to make a big piano designed to use a yoga mat as the base. For the keys, we used aluminum foil, which is both lightweight and conductive, allowing us to create a fun and interactive experience. To make it even more accessible, we incorporated conductive tape as the ground strip beneath the keys, enabling kids to play the piano using their feet without needing to hold anything to ground themselves. This design not only encourages movement and creativity but also allows for a unique way to engage with music. During our test trials, we observed how easily students could step on the keys to produce different sounds, and their enthusiasm was infectious.



Test #1: Trial one consisted of us trying to figure out how we can ground ourselves to the piano without having to use a strip on our hand, but we found out that if we use conductive tape as the ground strip beneath the keys, we will be able to play the piano using our feet without needing to hold anything to ground ourselves.

Test #2: For trial 2, we were trying to play the piano with our shoes on, but we found out that because of the rubber on of shoes, we were not able to ground the circuit. This is also a good place to talk about how rubber is used to insulate wires.

Test #3: For the third trial, we used our yoga mat as the base and alumiinum foil as the keys, with conductive tape as the ground stip beneath the keys, which allowed us to functionally play the piano on the ground with our feet when we connected the Makey Makey circuit board to the big screen.